

Question number	Answer	Additional guidance	Mark
7(a)(i)	7 (g) (1)		(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	- value given in the range 0.005 to 0.0083 (1) - this value given to 1 or 2 sig figs {per day / day⁻¹} (1)	ACCEPT answers in correct standard form to 1 or 2 sig figs ACCEPT {0.02 / 0.017} {per day / day⁻¹} for 1 mark	(2)

Question number	Answer	Additional guidance	Mark
7(a)(iii)	An explanation that includes four of the following points: • decomposition (of tea) is {faster / greater} at 25°C (1) • because enzymes {work faster / move faster / have more kinetic energy} (at warmer temperatures) (1) • as there are more (frequent / energetic) enzyme-substrate collisions (1) • loss of mass is due to release of carbon dioxide (1) • by the respiration of the {decomposers / bacteria / fungi} (1)	ACCEPT converse throughout ACCEPT description e.g. breakdown of organic matter ACCEPT may be due to more bacteria	(4)

Question number	Answer	Additional guidance	Mark
7(a)(iv)	An answer that includes the following points: - because the teas maybe composed of different molecules (1) - that {cannot be broken down as easily / are less accessible to the enzymes} (in rooibos) (1) OR - because the pH (in the teabag) is different (1) - therefore enzymes (of the bacteria) are less active (in rooibos) (1) OR - because there are {inhibitors / antimicrobials / toxins} in rooibos (1) - that {inhibit the enzymes / kill the decomposers} (1)	ACCEPT converse throughout ACCEPT named difference e.g. surface area IGNORE different masses of organic matter ACCEPT which enzymes cannot breakdown ACCEPT higher concentrations	(2)

Question number	Answer	Additional guidance	Mark
7(b)(i)	both the decomposition rate and {S / stabilisation factor / carbon stored} are mean values (1)	ACCEPT average for mean	(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes three of the following points: • {number 6 / loamy desert} (and 3 / birch / 5 / sandy desert) (1) • because it has the highest S value and the lowest decomposition rate (1) • therefore more carbon retained (in the soil) and less {carbon / carbon dioxide} released (1) • less carbon dioxide (in the atmosphere), the less global warming (1) OR • {number 6 / loamy desert} (and 3 / birch / 5 / sandy desert) (1) • because it has the highest S value therefore more carbon retained (in the soil) (1) • because it has the lowest decomposition rate and therefore less {carbon / carbon dioxide} released (1) • less carbon dioxide (in the atmosphere), the less global warming (1)	DO NOT ACCEPT carbon dioxide retained ACCEPT less greenhouse effect, description DO NOT ACCEPT carbon dioxide retained ACCEPT less greenhouse effect, description	(3)